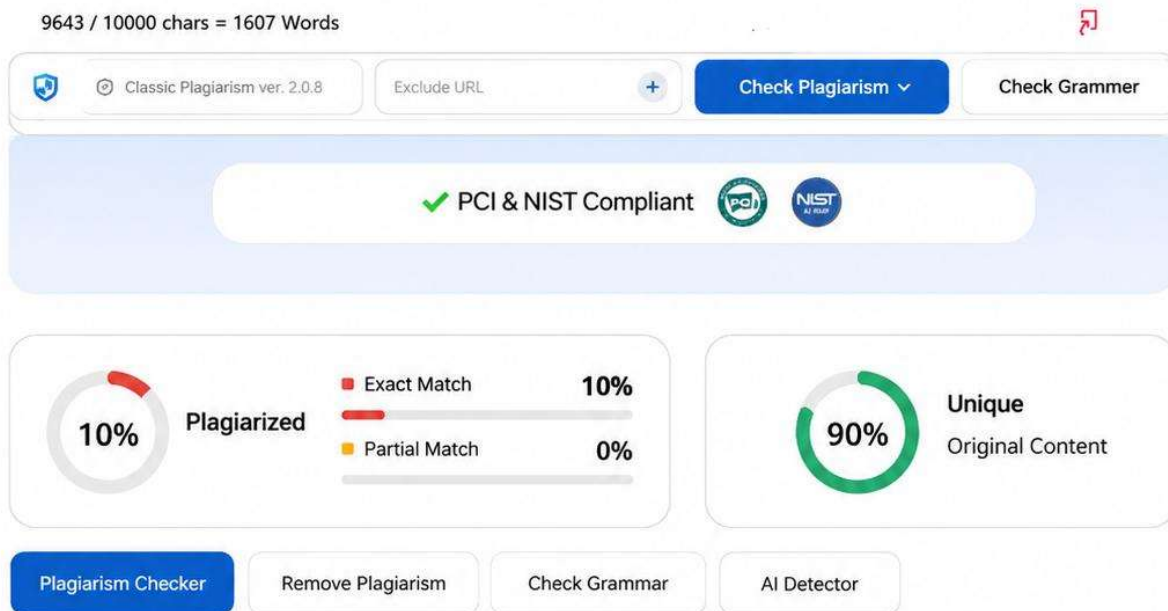


PLAGIARISM REPORT



Project: Student Skills Development Management System

Course: CSA4305 - Internet Programming

Degree: Bachelor of Engineering in Computer Science and Engineering

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PROJECT CONTENT

ABSTRACT

The Student Skills Development Management System is a web-based application designed to support students in identifying, developing, assessing, and monitoring their technical and soft skills. It provides a centralized platform for student profiles, career goals, skill assessments, skill-gap analysis, learning recommendations, progress tracking, achievements, and faculty monitoring.

A major feature is Skill Gap Analysis, which compares a student's current skill level with the required level for a selected career goal. The system can then highlight areas needing improvement and support continuous development through assessments, learning resources, progress indicators, charts, and reports.

1. PROJECT OBJECTIVES

1. Develop a secure student registration and authentication system.
2. Create and maintain individual student profiles.
3. Allow students to record technical and soft skills.
4. Conduct skill-based assessments and calculate performance scores.
5. Identify skill gaps by comparing current and required skill levels.
6. Provide recommendations and learning resources for weaker skills.
7. Track skill development using progress indicators and charts.
8. Provide an Admin/Faculty Dashboard for monitoring student development.

2. SCOPE

The system covers student and faculty authentication, student profile creation, career-goal selection, technical and soft skill management, skill assessments, automatic score calculation, skill-gap analysis, recommendations, learning-resource management, progress visualization, achievements, reports, and department-level analytics.

3. TOOLS AND TECHNOLOGIES

Category	Technology
Programming Language	Python

Frontend	HTML, CSS, JavaScript
UI Framework	Bootstrap
Backend	FastAPI
Database	MongoDB
Visualization	Chart.js
API Architecture	REST API
Server	Uvicorn
Development Environment	VS Code / Gravity
Category	Technology
Version Control	Git / GitHub

4. FUNCTIONAL MODULES

- User Authentication - secure registration, login, logout, and role-based access.
- Student Profile - stores personal, academic, career, and interest information.
- Skill Management - manages technical and soft skills.
- Skill Assessment - evaluates students using skill-specific questions.
- Score Calculation - converts assessment responses into percentage scores and proficiency levels.
- Skill Gap Analysis - compares current skill levels with target career requirements.
- Recommendation - prioritizes skills requiring improvement.
- Learning Resources - provides targeted learning content.
- Progress Tracking - represents development using progress bars and charts.
- Achievements - awards badges for milestones.
- Admin Dashboard - supports faculty monitoring.
- Report Generation - generates consolidated skill-development information.

5. ALGORITHM / METHODOLOGY

Skill Assessment: Load questions for a selected skill, evaluate each answer, calculate percentage = (score / total questions) x 100, classify the proficiency level, save the result, and display the score and level.

Skill Gap Analysis: Retrieve the student's career goal, current skills and required skills. For every required skill, calculate Skill Gap = Required Skill Level - Current Skill Level. Positive gaps are marked as needing improvement and are sorted to generate recommendations.

6. TESTING SUMMARY

Test Case	Expected Result	Status
Student Registration	Account created	PASS
Valid Login	Dashboard displayed	PASS
Invalid Login	Error displayed	PASS
Add / Edit / Delete Skill	Correct CRUD operation	PASS
Assessment	Score generated	PASS
Skill Gap	Correct gap calculated	PASS
Recommendation	Weak skills recommended	PASS
Course Progress	Progress updated	PASS
Achievement	Badge displayed	PASS
Admin Dashboard	Student information displayed	PASS

7. SDG RELEVANCE

SDG 4 - Quality Education: The project supports continuous learning by helping students identify skills, detect weaknesses, complete assessments, and monitor development. **SDG 8 - Decent Work and Economic Growth:** Career-oriented skill development can support preparation for internships, placements, entrepreneurship, and employment opportunities.

8. CONCLUSION

The Student Skills Development Management System integrates student profiles, skills, assessments, skill-gap analysis, recommendations, learning resources, progress tracking, analytics, and faculty monitoring in a single web-based platform. The project aims to improve transparency in student skill development and support academic and career preparation.